

Ayush Moharana

Research Associate at Keele University

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Observational astrophysicist studying eclipsing binaries and multiple stars. Expert in modelling of eclipsing binaries, eclipse timing variations, and stellar characterisation.

Experience

Research Associate

Keele University, UK
August 2024 - Present

Doctoral Candidate

Nicolaus Copernicus Astronomical Center, Poland
December 2019 - September 2024

Junior Research Fellow

National Institute of Technology, Rourkela, India
August 2019 - October 2019

Education

Ph.D. in Astronomy

Nicolaus Copernicus Astronomical Center, Poland
December 2019 – September 2024
Thesis: “Comprehensive study of low-mass compact hierarchical triples using eclipsing binaries”
Supervisor: Dr. hab. K. G. Helminiak

Integrated M.Sc. in Physics

National Institute of Technology, Rourkela, India
August 2014 – June 2019
Thesis: “Ultraviolet studies of galactic globular clusters using observations from Ultraviolet Imaging Telescope”
Supervisor: Dr. A.C. Pradhan

Publications

First Author

- “V446 Cephei: a β Cep pulsator in a multiple system”. A. Moharana et al. *MNRAS*, 2026, 548, stag772.
- “Formation of compact hierarchical triples”. A. Moharana et al. *Contributions of the Astronomical Observatory Skalnaté Pleso*, 2025, 55, pp. 280–285.
- “Spectroscopy of eclipsing compact hierarchical triples: I. Low-mass double-lined and triple-lined systems”. A. Moharana et al. *A&A*, 2024, 690, A153.
- “Solaris photometric survey: Search for circumbinary companions using eclipse timing variations”. A. Moharana et al. *MNRAS*, 2024, 527, pp. 53–65.
- “Detached eclipsing binaries in compact hierarchical triples: triple-lined systems BD+442258 and KIC 06525196”. Ayush Moharana et al. *MNRAS*, 2023, 521, pp. 1908–1923.

Second Author

- “RX Gru: A short-period pre-main-sequence eclipsing binary with a distant circumbinary companion”. F. Marcadon et al. *A&A*, 2026, 710, A394.
- “Adding TESS to CRÉME. Light curves and masses of 300+ eclipsing binaries”. K. G. Heminiak et al. *Contributions of the Astronomical Observatory Skalnaté Pleso*, 2025, 55, pp. 438–443.
- “Discovery of a hot post-AGB star in Galactic globular cluster E3”. R. Kumar et al. *A&A*, 2024, 685, p. L6.
- “Orbital and physical parameters of eclipsing binaries from the ASAS catalogue - XII. A sample of systems with K2 photometry”. K. G. Heminiak et al. *MNRAS*, 2021, 508, pp. 5687–5708.
- “Consequences of parameterization choice on eclipsing binary light curve solutions”. J. Korth et al. *Contributions of the Astronomical Observatory Skalnaté Pleso*, 2021, 51, pp. 58–67.

Contributing Author

- “Time-Domain Photometry and Activity Evolution of Interstellar Comet 3I/ATLAS with BHTOM”. A. Fraser Gillan et al. *arXiv e-prints*, 2026, arXiv:2603.01383.
- Pierre F. L. Maxted et al. *Detached eclipsing binary star science in the 2040s*. Jan. 2026. arXiv: 2601.17179 [astro-ph.IM].
- “A comprehensive study of δ Scuti-type pulsators in eclipsing binaries: Oscillating eclipsing Algols”. T. Pawar et al. *A&A*, 2026, 705, A251.
- “A comprehensive study of five candidate δ Scuti-type pulsators in detached eclipsing binaries”. T. Pawar et al. *A&A*, 2024, 691, A101.
- “Comprehensive spectroscopic and photometric study of pulsating eclipsing binary star Al Hya”. F. Kahraman Aliçavu et al. *MNRAS*, 2023, 520, pp. 1601–1612.
- “The Cluster Ages Experiment (CASE) - IX. Analysis of four detached eclipsing binaries in the globular cluster NGC 3201”. M. Rożyczka et al. *MNRAS*, 2022, 517, pp. 2485–2501.
- “Ultraviolet Imaging Telescope (UVIT) observation of the Galactic globular cluster NGC 7492”. Ranjan Kumar et al. *MNRAS*, 2021, 502, pp. 313–327.

Conference Proceedings

- “Multiplicity of Cool Stars and their Evolution”. Matthew I. Swayne et al. *CoolStars 23, Tokyo*, 2026, arXiv:2609.02400.
- “Predicting the fate of Compact Hierarchical Triples: From precise parameters to stellar and orbital evolution”. Ayush Moharana and Krzysztof Heminiak, 2023, p. 1434.
- “Observational constraints on the evolution and dynamics of Compact Hierarchical Triples”. Ayush Moharana and Krzysztof Heminiak, 2023, p. 855.
- “Observational probe into evolution of Compact Hierarchical Triples”. Ayush Moharana et al., 2023, p. 46.
- “Evolution and Dynamics of Tight Triple Systems”. Ayush Moharana et al., 2022, vol. 12, pp. 198–201.
- “Accurate stellar parameters for stars in Tight Triple Systems”. Ayush Moharana et al., 2021, p. 51.

Presentations, Seminars, and Talks

- *Flash talk* at **Cool Stars 23**, Tokyo, 2026
- *Contributed talk* at **Binary Stars in Space Era**, Keele, 2025
- *Contributed talk* at **Binary and Multiple Stars in the Era of Big Sky Surveys**, Litomysl, September, 2024
- *Invited seminar* at **Nicolaus Copernicus Astronomical Center**, Torun, Poland, April, 2024
- *Contributed talk* at **International Meeting on Eclipsing Binary Star System**, Paris, November, 2023
- *Invited seminar* at **Journal Club, Princeton University**, Princeton, United States, October, 2023
- *Contributed talk* at **Polish Astronomical Society meeting**, Torun, September 2023
- *Contributed talk* at **European Astronomical Society meeting**, Krakow, July 2023.
Session: LS11- Evolving Observations of Binary Stars
- *Contributed talk* at **European Astronomical Society meeting**, Krakow, July 2023.
Session: S11- Stellar interactions
- *Contributed talk* at **Alpha Cen Systems: Towards new worlds**, Nice, June 2023
- *Contributed talk* at **Science highlights from SALT**, Warsaw, June 2023
- *Flash talk* at **Impact of Binaries on Stellar Evolution (ImBaSE)**, Munich Institute for Astro, Particle and Bio-physics, Munich, Germany, November 2022.
- *Contributed talk* at **STARS 2020**, Institute of Astronomy, University of Cambridge, UK. August 2022.

Awards, Grants, and Fellowships

- Young Researcher of the Year 2023, Nicolaus Copernicus Astronomical Center – *Awarded for the multidimensional research on triple star systems, specifically for the integration of theoretical studies with observations.*
- PRELUDIUM Research Grant, National Science Center, Poland – *3-year grant of 203,096 PLN for the project 'Evolution of tight triple stars'.*

- ISRO project fellowship, Indian Space Research Organisation – *1-year fellowship to work on the project 'Constraining Milky Way structure using UV star counts'.*
- Best poster award at “Stellar Variability: Taking the Pulse of the Universe”, Pune, 2025 – *Poster title: Insights into tidally perturbed β Cep stars in eclipsing binaries.*

Observing programmes

- **PI, SALT 4-semester monitoring with HRS**, 2024-2026 – *'Age and metallicity dependency of orbital parameters of compact hierarchical triples'.*
- Co-I, SALT HRS spectroscopy, 2024 – *Detached Eclipsing Binaries as Benchmark Stars for PLATO mission*
- **PI, 60 hrs CHIRON spectroscopy with CTIO**, 2023
- **PI, SALT 4-semester monitoring with HRS**, 2022-2024 – *Accurate parameters of stars in Tight Triple Systems'.*
- Co-I, TESS Guest Investigator programmes, 2020, 2021, 2022

Observing and Technical Experience

- Observer at the Danish 1.54m telescope, La Silla Observatory, Chile – 28 nights
- Developer of the photometric pipeline for the *Solaris* network of 0.5m robotic telescopes.

Teaching Experience

- Teaching assistant, Astrophysics project laboratory, Keele University – Autumn, 2024 and 2025.

Organisation and Service

- Reviewer for Nature Portfolio Journals.
- Reviewer for European national grants.
- Web administrator, Stellar Multiplicity Database: <https://stellarmultiplicity.github.io/database/>
- Chair, scientific organising committee, session on “Binary stars: Current developments and preparing for future missions”, National Astronomy Meeting, Birmingham, UK, July 2026.
- Co-chair, scientific organising committee, splinter session on “Multiplicity of cool stars and their evolution”, CoolStars23, Tokyo, Japan, June 2026.
- Co-chair, local organising committee, Binary stars in the space era, Keele, UK, June 2025.
- Member, local organising committee, Polish Astronomical Society Meeting, Torun, Poland, September 2023.

Public Outreach

- “Tatooine and other weird worlds”, Pint of Science Festival, Stoke-on-Trent, May 2026.
- Talk at Young Astronomers meet, Warsaw, March 2022.
- Talk on “Measuring Stars with Eclipsing Binaries at Astronomy Students Scientific Association, NCU Torun, May 2021.
- Popular science talk on “Multiple Star systems” at National Institute of Technology, Rourkela, 2020.
- Astronomy workshop for rural kids, Chandbali, November 2019.

Other skills

Graphic design, Diploma in fine arts, Amateur tennis player

References

Dr. J. Southworth (PostDoc supervisor)

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🏠 Keele University, United Kingdom

Dr. hab. K.G. Heminak (PhD supervisor)

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🏠 Nicolaus Copernicus Astronomical Center, Polish Academy of Sciences, Torun, Poland